FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

AARHUS
UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT

TLDR

Two hours finding out what actuators feel like, and leaving able to drive one from your own code.

Get a board and run task 01. Something spins.

Change two numbers, upload again. That loop is the workshop.

Pick up the actuators you are not driving.

Work through as many of the five tasks as you get to.

Build one signal someone else can read without looking.

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew the lab has 117 coin motors

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

A board each. You build the circuits yourself

- Every part, every wire, from an empty breadboard

Four tasks, each one wire more than the last

- Piezo, then a touch pad, then both, then a stepper

Actuators are on the tables at the front

- Go and feel the ones you did not build

Finish by building one signal someone else can read

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Task 00: piezo. Two wires, everyone
0:25	20 min	Task 04: one wire and foil. Watch the drift
0:45	20 min	Task 05: join them. Touch in, tick out
1:05	20 min	Task 07: stepper, if you want it
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down

THE TASKS, IN BUILD ORDER

Each one adds a single new thing to the circuit before it.

00	**Make it tick**	**piezo**	**2 wires**
04	**Make it notice you**	**a wire and foil**	**1 wire**
05	**Make it answer**	**both of the above**	**3 wires**
07	**Make it turn**	**stepper**	**6 wires**

Four tasks, and the first three need no driver at all.

Tasks 01 to 03 are in the repo if you want the motor,

the solenoid or the Peltier tile. We will not do them today.

HOW THE TASK SKETCHES WORK

Wire it from the diagram at the top, then upload. All of them: github.com/labtools-au/multimodal-actuator-workshop

A PINS block at the very top

- Every leg, every wire. Build from that, not from memory

A CHANGE ME block below it

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments. The last is the interesting one

PIN MAP

01	spin	D9 PWM	1k to PN2222 base, motor on collector
02	tap	D6	MOSFET gate, own supply, grounds joined
03	warm	D3 + D5 PWM	H-bridge low and high, bench supply
04	sense	A0	bare wire to foil. That is all of it
05	answer	A0 + D9	the pad, plus any actuator
07	stepper	D8 D9 D10 D11	ULN2003 IN1 to IN4

Only D3, D5, D6, D9, D10, D11 do PWM. Stepper pins go into the constructor OUT of order: 8, 10, 9, 11.

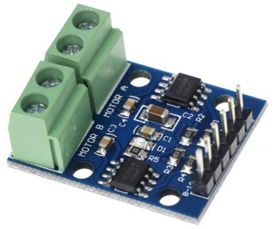
WHAT YOU GET IN THE BASE KIT

One set per pair, collected before you start task 01. Below: the three parts people always come back for.

Board, breadboard, jumper wires, resistors: on the tables.

Not in the drawers, so take what you need.

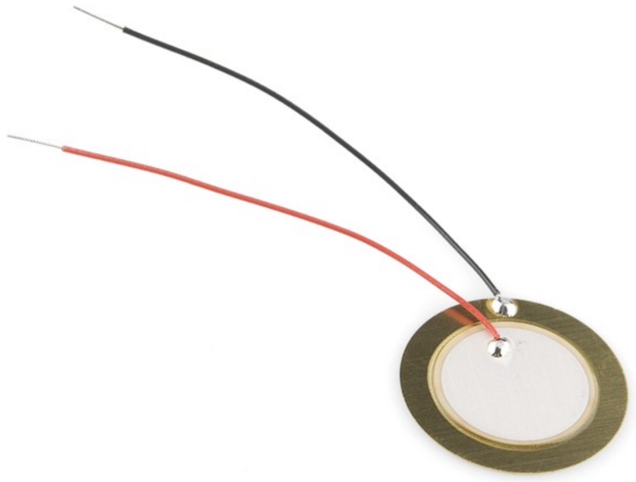
Breadboard supply, 8A. Anything past an LED.



L9110S H-bridge, 2C. Reversing, and the Peltier.

PAM8403 amp, 11F. A transducer needs one.

PIEZO ELEMENT



A crystal that flexes when you put current across it. Near-instant, almost no power, but shallow.

Feels like: a tick, not a thump.

USED FOR

Clicks and confirmations. Anything where lag would feel broken.

`Drawer 6F | 69 in stock | two wires, no driver`

CAPACITIVE PAD

PHOTO
Capacitive pad

A wire taped behind foil, card or fabric. No sensor and no breakout board, which is why the box on the left is empty.

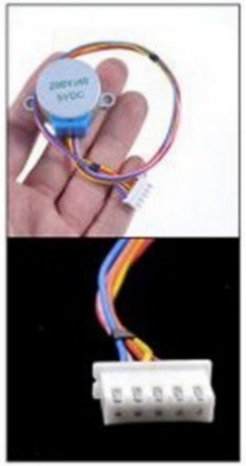
Feels like: nothing. That is the point. The surface stays plain.

USED FOR

Invisible controls in wood or fabric, waterproof panels.

`No drawer, no part number, no cost | one wire | keep it on USB`

STEPPER MOTOR



Moves in fixed steps rather than continuously, so you always know where it is without a sensor.

Feels like: precise, and audibly clicky.

USED FOR

Slow accurate motion. Dials, sliders, anything positioned.

`2A | 44 | ULN2003 | 2048 steps/rev | own 5V supply`

WHAT ELSE THE LAB HAS

You are not limited to today's three. All of this is in the drawers, and the repo has working code for the first three.

Coin motor a buzz you cannot make gentle and fast

- 1E | 117 in stock | needs a transistor

Solenoid a person tapping you, not a signal

- 3E | 24 | 1.1 A, needs its own supply

Peltier tile heat and cold, several seconds to be sure

- 4C | 60 | needs an H bridge

Surface transducer turns any surface into a speaker

- 7F | 48 | needs an amplifier

Servo holds an angle, with force behind it

- 2F | 80 | Servo.h, own 5V

FREESTYLE, WITHIN THREE LIMITS

You do not have to do the tasks in order, or finish them. Playing properly with two beats rushing all five.

If you want to drive something not in front of you, go get it.

The three things that are not negotiable

• Solenoid and Peltier get their own supply, never USB

• The Peltier heatsink goes on before it is switched on

• No motor straight from a pin. 40 mA limit, it wants 75

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

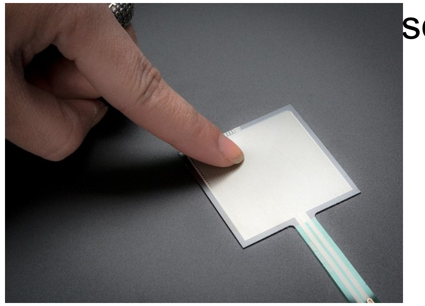
INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- GSR, drawer 0C5. Exactly one in the lab, so ask first

so



Force (FSR), 4A. 46 tiny, 33 square. Calibrate each one.

Flex, 4B. Only 8 left, so share them.

Pulse, 9D. 9 in stock. Steady at rest, useless once moving.

GUSTAV SIMONSEN
TEACHING ASSISTANT

YOU ALREADY WRITE CODE. WHY COME?

Fair question. Three honest answers.

You cannot Google what something feels like

- No datasheet says a knock reads as a person, a buzz as a machine

The parts are free and already here

- Borrowing beats ordering. Nothing to buy, nothing to wait for

Your devices hide sensors you may not be able to reach

- Find out today which open up, and which are locked

REVERSE ENGINEER WHAT YOU ALREADY OWN

The trackers in your pocket and on your wrist. What opens up, and what does not.

BLE heart rate strap: readable from a web page

• Web Bluetooth, standard GATT. Chrome and Edge only

Most Garmin watches: the same, with no app

• Turn on Broadcast Heart Rate and it acts as a strap

Apple Watch: nothing live, whatever you write

• HealthKit needs a watchOS app in Swift

Garmin history: Connect IQ, or the API afterwards

• Good for analysis. Useless for reacting in real time

WHAT A PHONE WILL AND WILL NOT GIVE YOU

Open sensors.chomskylab.dk on your own phone right now.

Works in a web page

- Accelerometer and gyroscope, microphone, camera
- Pointer pressure, though a trackpad reports only 0 or 0.5

sensors.chomskylab.dk

Does not, whatever the tutorial says

- Phone vibration on iOS. WebKit has never shipped it, at any version, and Chrome on iOS is Safari underneath

So "the phone buzzes" works on no iPhone in this room.

- Check the feature, not the tutorial, before it is in a plan.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator, whichever you built

- Three messages: arrived, something wrong, finished

Vary rhythm above all

- Piezo: rhythm is all you have. Motor: strength too

Blind test: they name all three

- Note which two got confused, and why

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,

you built a visual interface with a motor glued to it.

BEFORE YOU CALL IT DONE

Hand your device to another group with the screen turned away.

- They can tell all three messages apart
- It does not fire continuously when held
- You know which two got confused, and why

What you actually leave with

- What things feel like, how to drive one, and which to pick
- That last one is the expensive mistake this prevents

GO AND TOUCH THINGS

Everything is open: code, slides and session plan

github.com/labtools-au/
multimodal-actuator-workshop

GUSTAV SIMONSEN
TEACHING ASSISTANT